

Proposition 5.1

parabolic rescaling		Han--Li--Sun	Proposition 5.1		(X_0, T)	
Andrews--Chow--Guenther--Langford		<i>Extrinsic Geometric Flows</i>		369		
$X_j(x, t) = \lambda_j \Big(X(x, t_j + \lambda_j^{-2} t) - X(x_j, t_j) \Big), \qquad \lambda_j = H(x_j, t_j).$						
Proposition 5.1		spacetime L^2				
mean curvature flow		shrinking sphere		$ H_\infty (0, 0) = 1$	smooth pointed	
blow-up limit						
Han--Li--Sun		beta-symplectic gradient flow			fixed-	
center Proposition 5.1						
$q_j = \lambda_j \big(F(x_j, t_j) - X_0 \big), \qquad \theta_j = \lambda_j^2 (T - t_j).$						
q_j	θ_j	H_j, V_j, f_j	L^2	$F_j^\perp$	$q_j^\perp$	L^2
$q_j \rightarrow 0$		spacetime area bound		Han--Li--Sun moving-cutoff		
(5.5)						
				$F(x_j, t_j) \rightarrow X_0$	$q_j \rightarrow 0$	
$ F(x_j, t_j) - X_0 = o(\lambda_j^{-1});$						
essential Type-I		$\lambda_j \asymp (T - t_j)^{-1/2}$		$o(\sqrt{T - t_j})$	$\lambda_j = H , A , (T -$	
$t_j)^{-1/2}$		q_j	$q_j \rightarrow 0$		Chen--Li	
4.7	q_j	x_k		p	cutoff	
$q = 0$						

1.

$F : \Sigma \times [0, T) \rightarrow M$	beta-symplectic critical-surface gradient flow	normal coordinates
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$$\partial_\tau F = f, \quad f = \frac{\cos^2 \alpha H - \beta \sin^2 \alpha V}{\cos^2 \alpha + \beta \sin^2 \alpha},$$

H mean-curvature vector V Han--Li--Sun normal field α Kähler angle
symplectic-angle bound

$$\cos \alpha \geq \delta > 0.$$

Han--Li--Sun (X_0, T)

$$G_\lambda(x,u) = \lambda \big(F(x, T + \lambda^{-2}u) - X_0 \big), \quad u < 0.$$

Proposition 5.1 $R > 0$ $-\infty < a < b < 0$ $\lambda \rightarrow \infty$

$$\begin{aligned} \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |H_\lambda|^2 d\mu_u^\lambda du, \quad \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |V_\lambda|^2 d\mu_u^\lambda du, \\ \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |f_\lambda|^2 d\mu_u^\lambda du, \quad \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |G_\lambda^\perp|^2 d\mu_u^\lambda du. \end{aligned}$$

369 $x_j \in \Sigma \quad t_j \nearrow T$

$$p_j = F(x_j, t_j), \quad \lambda_j = |H|(x_j, t_j) \longrightarrow \infty,$$

$$F_j(x,t) = \lambda_j \Big(F(x, t_j + \lambda_j^{-2}t) - p_j \Big). \tag{1.1}$$

rescaled time interval

$$F_j(x_j,0)=0, \qquad |H_j|(x_j,0)=1. \tag{1.2}$$

(1.1) $R > 0 \quad s_1 < s_2 < 0$ Proposition 5.1

2. scaling identities

$$\tau_j(t) = t_j + \lambda_j^{-2}t.$$

translation tangent normal spaces ambient dilation $y \mapsto \lambda_j y$ induced metric λ_j^2 Σ

$$g_j = \lambda_j^2 g, \quad d\mu_t^j = \lambda_j^2 d\mu_{\tau_j(t)}, \quad d\tau = \lambda_j^{-2} dt. \tag{2.1}$$

second fundamental form inverse metric

$$H_j = \lambda_j^{-1} H. \tag{2.2}$$

Kähler angle dilation scaled orthonormal tangent frame λ_j^{-1}

$$V_j = \lambda_j^{-1} V. \quad (2.3)$$

$$f_j = \frac{\cos^2 \alpha_j H_j - \beta \sin^2 \alpha_j V_j}{\cos^2 \alpha_j + \beta \sin^2 \alpha_j} = \lambda_j^{-1} f, \quad \partial_t F_j = f_j. \quad (2.4)$$

$\cos \alpha_j \geq \delta$ uniform pointwise estimate

$$|f_j|^2 \leq C(\beta, \delta) (|H_j|^2 + |V_j|^2). \quad (2.5)$$

fixed singular point (X_0, T)

$$q_j = \lambda_j(p_j - X_0), \quad \theta_j = \lambda_j^2(T - t_j). \quad (2.6)$$

fixed-center coordinates

$$G_j(x, t) := \lambda_j(F(x, \tau_j(t)) - X_0) = F_j(x, t) + q_j, \quad (2.7)$$

$$T - \tau_j(t) = \lambda_j^{-2}(\theta_j - t). \quad (2.8)$$

backward heat kernel

$$\rho_{X_0, T}(F, \tau_j(t)) d\mu_{\tau_j(t)} = \psi_j(F_j, t) d\mu_t^j, \quad (2.9)$$

$$\psi_j(z, t) = \frac{1}{4\pi(\theta_j - t)} \exp\left(-\frac{|z + q_j|^2}{4(\theta_j - t)}\right). \quad (2.10)$$

normal-position identity

$$G_j^\perp = (F_j + q_j)^\perp = F_j^\perp + q_j^\perp. \quad (2.11)$$

fixed-center monotonicity

$$(F_j + q_j)^\perp \quad F_j^\perp$$

3. Gaussian lower bound sharp criterion

$R > 0$ $s_1 < s_2 < 0$ cylinder

$$|F_j| \leq R, \quad s_1 \leq t \leq s_2$$

j $c_R > 0$ $\psi_j \geq c_R$

$$\sup_j |q_j| < \infty, \quad \sup_j \theta_j < \infty. \quad (3.1)$$

4.2 smooth nonflat limit obstruction

F_∞ round symmetry moving-center rescalings smooth locally converges
normalization

$$|H_\infty|(p_\infty, 0) = 1.$$

compact coordinate patch K $s_1 < s_2 < 0$

$$|H_\infty| \geq \frac{1}{2} \quad \text{on } K \times [s_1, s_2].$$

smooth convergence j $|H_j| \geq 1/4$ induced area density patch R
 $F_j(K, t) \subset B_R(0)$ $c > 0$

$$\liminf_{j \rightarrow \infty} \int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 d\mu_t^j dt \geq c > 0. \quad (4.2)$$

beta-symplectic curvature-normalizing sequence smooth nonflat blow-up limit

$$|H_j|(x_j, 0) = 1 \quad \text{compactness} \quad (4.2)$$

regularity

5. beta-symplectic flow

Han--Li--Sun fixed-center Proposition 5.1

Theorem 5.1 moving-center transfer

F beta-symplectic gradient flow (X_0, T) fixed singular point $\lambda_j \rightarrow \infty$ (1.1) (2.6)
 F_j, q_j, θ_j

1. λ_j fixed-center rescalings

$$\tilde{F}_j(x, u) = \lambda_j (F(x, T + \lambda_j^{-2} u) - X_0)$$

Han--Li--Sun Proposition 5.1 fixed-center spacetime L^2

2. $\sup_j |q_j| < \infty$ $0 \leq \theta_j \leq \Theta < \infty$

3. $[s_1, s_2] \subset (-\infty, 0)$ j moving rescaled time domain

$R > 0$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 d\mu_t^j dt \rightarrow 0, \quad (5.1)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |V_j|^2 d\mu_t^j dt \rightarrow 0, \quad (5.2)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |f_j|^2 d\mu_t^j dt \rightarrow 0. \quad (5.3)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |F_j^\perp|^2 d\mu_t^j dt \rightarrow 0, \quad (5.4)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |q_j^\perp|^2 d\mu_t^j dt \rightarrow 0. \quad (5.5)$$

$$(5.5) \quad \text{normal-displacement condition} \quad \text{Han--Li--Sun} \quad \text{moving-cutoff} \quad (5.5)$$

$$q_j \rightarrow 0 \quad (5.6)$$

$$A_R < \infty$$

$$\int_{s_1}^{s_2} \mu_t^j(\Sigma_t^j \cap B_R(0)) dt \leq A_R, \quad (5.7)$$

$$(5.4) \quad \text{moving-center}$$

Proof

moving time t

$$u = t - \theta_j.$$

$$\theta_j = \lambda_j^2(T - t_j)$$

$$T + \lambda_j^{-2}u = t_j + \lambda_j^{-2}t.$$

$$\widetilde{F}_j(x, u) = F_j(x, t) + q_j. \quad (5.8)$$

$$\text{translation} \quad H_j, V_j, f_j \quad \text{normal spaces} \quad \text{induced measure} \quad |F_j(x, t)| < R \quad |q_j| \leq Q$$

$$|\widetilde{F}_j(x, u)| < R + Q.$$

$$0 \leq \theta_j \leq \Theta \quad t \in [s_1, s_2]$$

$$u \in [s_1 - \Theta, s_2] \subset (-\infty, 0).$$

integrands

$$\int_{s_1}^{s_2} \int_{\{|F_j| < R\}} |H_j|^2 \leq \int_{s_1 - \Theta}^{s_2} \int_{\{|\widetilde{F}_j| < R + Q\}} |\widetilde{H}_j|^2.$$

$$\begin{array}{ccccccc} \text{fixed-center Proposition 5.1} & (5.1) & \text{inclusion} & \text{time shift} & (5.2) & (5.3) \\ (2.5) & (5.1) & (5.2) & & & (5.3) \end{array}$$

$$\text{normal-position term} \quad (2.11)$$

$$\widetilde{F}_j^\perp = F_j^\perp + q_j^\perp.$$

$$\text{fixed-center Proposition 5.1} \quad \text{ball/time inclusion}$$

$$\int_{s_1}^{s_2} \int_{\{|F_j| < R\}} |F_j^\perp + q_j^\perp|^2 \rightarrow 0. \tag{5.9}$$

$$|F_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|q_j^\perp|^2$$

$$(5.5) \Rightarrow (5.4)$$

$$|q_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|F_j^\perp|^2$$

$$(5.9) \quad (5.4) \Rightarrow (5.5) \quad \text{orthogonal projection}$$

$$\int |q_j^\perp|^2 \leq |q_j|^2 \int d\mu_t^j dt.$$

$$(5.6) \quad (5.7)$$

6.

$$\begin{array}{ccccc} q_j & \theta_j & \text{shifted normal-position limit} & F_j^\perp \text{ limit} & \text{affine plane} \end{array}$$

$$F_j(u,v,t) = (u,v,-1) \subset \mathbb{R}^3, \quad q_j = (0,0,1).$$

$$H_j = V_j = f_j = 0$$

$$F_j + q_j = (u,v,0)$$

$$\text{tangent} \quad (F_j + q_j)^\perp = 0$$

$$F_j^\perp = (0,0,-1).$$

$$R > 1 \quad s_1 < s_2$$

$$\int_{s_1}^{s_2} \int_{F_j \cap B_R(0)} |F_j^\perp|^2 d\mu dt = (s_2 - s_1)\pi(R^2 - 1) > 0.$$

$$\text{center displacement} \quad \text{normal component} \quad \quad \quad \text{“} \quad \quad \quad \text{”}$$

7.

$$q_j \rightarrow 0$$

$$X_0 \quad \text{normal coordinates}$$

$$d_j = |F(x_j, t_j) - X_0|, \quad q_j = \lambda_j(F(x_j, t_j) - X_0). \quad (7.1)$$

Proposition 7.1

$$t_j \nearrow T \quad d_j \rightarrow 0 \quad \lambda_j > 0$$

$$q_j \rightarrow 0 \quad \Longleftrightarrow \quad d_j = o(\lambda_j^{-1}). \quad (7.2)$$

$$|A|^2(x, t) \leq \frac{K}{T-t}, \quad (7.3)$$

$$\begin{aligned} \lambda_j &= |A|(x_j, t_j) \quad \lambda_j = |H|(x_j, t_j) \\ d_j &= o(\sqrt{T-t_j}) \end{aligned} \quad (7.4)$$

$$\begin{aligned} q_j \rightarrow 0 \quad \text{essential Type-I} \quad c > 0 \\ \lambda_j \sqrt{T-t_j} \geq c, \end{aligned} \quad (7.5)$$

$$(7.4) \quad \lambda_j = (T-t_j)^{-1/2} \quad (7.4) \quad q_j \rightarrow 0$$

Proof

$$|q_j| = \lambda_j d_j. \quad (7.6)$$

$$(7.2) \quad \text{Cauchy--Schwarz}$$

$$|H|^2 \leq 2|A|^2. \quad (7.7)$$

$$(7.3) \quad \lambda_j = |A| \quad \lambda_j \sqrt{T-t_j} \leq \sqrt{K} \quad \lambda_j = |H| \quad \lambda_j \sqrt{T-t_j} \leq \sqrt{2K} \quad (7.6)$$

$$\begin{aligned} |q_j| &= (\lambda_j \sqrt{T-t_j}) \frac{d_j}{\sqrt{T-t_j}}, \\ (7.5) \quad |q_j| &= d_j / \sqrt{T-t_j} \end{aligned}$$

7.2

$$\begin{aligned} o \quad d_j \rightarrow 0 \quad T = 1 \\ F : S^2 \times [0, 1) \longrightarrow \mathbb{R}^4, \quad F(u, t) = 2\sqrt{1-t}u, \end{aligned} \quad (7.8)$$

$$S^2 \quad \mathbb{R}^4 \quad \partial_t F = H$$

$$|A|^2=\frac{1}{2(1-t)},\qquad |H|=\frac{1}{\sqrt{1-t}}. \tag{7.9}$$

$$t=1 \qquad u_0\in S^2 \qquad x_j=u_0 \quad t_j\nearrow 1 \quad X_0=0 \qquad x_j$$

$$F(x_j,t_j)=2\sqrt{1-t_j}\,u_0\longrightarrow 0.$$

$$\begin{array}{c|c} \lambda_j & q_j=\lambda_j F(x_j,t_j) \\ \hline |H|(x_j,t_j) & 2u_0 \\ |A|(x_j,t_j) & \sqrt{2}\,u_0 \\ (1-t_j)^{-1/2} & 2u_0 \end{array} \tag{7.10}$$

$$\begin{array}{ccc} \text{mean curvature flow} & \text{Han--Li--Sun} & \text{beta-symplectic flow} \\ \text{``} & q_j\rightarrow 0\text{''} & \text{beta-symplectic} \\ & (7.2) & o \end{array}$$

7.3

$$\begin{array}{ccccc} q_j\rightarrow 0 & & (7.4) & & d_j=o(\lambda_j^{-1}) \\ q_j\rightarrow 0 \quad \text{Chen--Li} & & q_j & & q_j\rightarrow q \end{array}$$

8. Chen--Li **4.7**

$$\text{Chen--Li} \qquad 305\text{--}308 \qquad \text{K\"ahler--Einstein} \qquad \text{symplectic surface} \qquad \text{mean curvature flow}$$

$$\lambda_k^2=|A|^2(x_k,t_k)=\max_{t\leq t_k}|A|^2,\qquad x_k\rightarrow p,\quad t_k\nearrow T, \tag{8.1}$$

$$F(p,T)=X_0=0$$

$$F_k(x,t)=\lambda_k\Big(F(x,t_k+\lambda_k^{-2}t)-F(p,t_k)\Big). \tag{8.2}$$

$$\mathbf{8.1} \qquad q_k \qquad q_k\rightarrow 0$$

$$|H|\leq \sqrt{2}|A|$$

$$|F(p,t_k)-F(p,T)|\leq \int_{t_k}^T |H(p,t)|\,dt\leq C\sqrt{T-t_k}. \tag{8.3}$$

$$\lambda_k\sqrt{T-t_k}\leq C$$

$$|q_k|:=|\lambda_k F(p,t_k)|\leq C. \tag{8.4}$$

$q_k \rightarrow q$
 $\lambda_k F(p, t_k) \rightarrow q$

$q = 0$
 q

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4.7

“we assume that

8.2

(8.1)--(8.2)

$|A_k|^2(x_k, 0) = 1,$
 $F_k(x_k, 0) = \lambda_k(F(x_k, t_k) - F(p, t_k)).$
(8.5)

p

$|A_k|^2(p, 0) = \lambda_k^{-2}|A|^2(p, t_k).$
(8.6)

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 $|A_k|^2(p, 0) \geq c > 0$
 $x_k \rightarrow p$
 $|A|^2(p, t_k) \geq c\lambda_k^2$
 p
 x_k
 $d(x_k, p)$
(8.6)
 p
ambient ball

$\lambda_k|F(x_k, t_k) - F(p, t_k)| \leq C$
(8.7)

(8.6)
(8.7)
smooth-limit
 $|A_\infty|(0, 0) > 0$

8.3
cutoff

$s = t_k + \lambda_k^{-2}t$
 $P_k = F(p, t_k)$
 $d\mu_t^k = \lambda_k^2 d\mu_s$
 $F_k + \lambda_k P_k = \lambda_k F,$
 $t_k - s = -\lambda_k^{-2}t.$

cutoff

$$\int_{\Sigma_t^k} \frac{1}{v} \phi_R(F_k) \frac{1}{-t} \exp\left(-\frac{|F_k + \lambda_k P_k|^2}{4(-t)}\right) d\mu_t^k$$

$$= \int_{\Sigma_s} \frac{1}{v} \phi_R(\lambda_k(F - P_k)) \frac{1}{t_k - s} \exp\left(-\frac{|F|^2}{4(t_k - s)}\right) d\mu_s.$$
(8.8)

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 $\phi_R(F)$
cutoff

$\chi \in C_c^\infty$
 $\chi = 1$
 X_0
 $\lambda_k P_k$
rescaled

ball $|Y| \leq R$

$F = P_k + \lambda_k^{-1}Y \longrightarrow X_0$
(8.9)

(8.8)
 k
 $\chi(F) \equiv 1$
fixed original-coordinate cutoff
 $\phi_R(F_k)$
 k -dependent moving cutoff

8.4

monotonicity
compactness
smooth immersed surface $F_\infty : \Sigma_\infty \rightarrow \mathbb{R}^4$

$$H_\infty = 0, \quad (F_\infty + q)^\perp = 0. \quad (8.10)$$

$$\Omega = \{F_\infty + q \neq 0\} \quad F_\infty + q \quad \text{tangent vector} \quad F_\infty + q = dF_\infty(W) \quad \text{tangent vector } X$$

$$D_X(F_\infty + q) = dF_\infty(X)$$

$$\text{normal component Gauss} \quad A_\infty(X, W) = 0 \quad e_1 = W/|W|$$

$$A_\infty(e_1, e_1) = A_\infty(e_1, e_2) = 0.$$

$$H_\infty = A_\infty(e_1, e_1) + A_\infty(e_2, e_2) = 0 \quad A_\infty(e_2, e_2) = 0 \quad A_\infty = 0 \quad \Omega \quad \Omega \quad F_\infty + q$$

$$F_\infty \quad \text{immersion} \quad A_\infty \equiv 0$$

$$q \quad q = 0$$

8.5

$$\text{compactness} \quad \text{monotonicity} \quad q = 0$$

Theorem 8.1 q **blow-up contradiction** . $G_k : M_k \rightarrow \mathbb{R}^N$ rescaled time-slices
 $p_k \in M_k$

$$|A_k|(p_k) = 1, \quad q_k \rightarrow q \in \mathbb{R}^N, \quad \theta_k \rightarrow \theta \in (0, \infty). \quad (8.11)$$

$$U \quad p_\infty \in U \quad \text{compact exhaustion } K_i \Subset U \quad K_i \quad p_\infty \quad \text{embeddings}$$

$$\Phi_{k,i} : K_i \rightarrow M_k$$

$$\Phi_{k,i}(p_\infty) = p_k, \quad G_k \circ \Phi_{k,i} \longrightarrow G_\infty \quad \text{in } C^2(K_i). \quad (8.12)$$

$$i \quad \varepsilon \in \{-1, 1\} \quad \text{localized estimates}$$

$$\int_{K_i} |H_k|^2 d\mu_k \longrightarrow 0, \quad (8.13)$$

$$\int_{K_i} \left| H_k + \frac{\varepsilon}{2\theta_k} (G_k + q_k)^\perp \right|^2 d\mu_k \longrightarrow 0, \quad (8.14)$$

$$\text{contradiction} \quad q_k \quad q = 0$$

Proof. (8.12) induced metrics area densities mean curvature vectors normal projections K_i
(8.13)--(8.14)

$$H_\infty = 0, \quad H_\infty + \frac{\varepsilon}{2\theta} (G_\infty + q)^\perp = 0 \quad \text{on } K_i^\circ. \quad (8.15)$$

$$K_i^\circ \quad U \quad \theta > 0 \quad H_\infty = 0 \quad (G_\infty + q)^\perp = 0 \quad U \quad 8.4 \quad A_\infty \equiv 0 \quad (8.11)\text{--}(8.12) \quad C^2 \text{ convergence}$$

$$|A_\infty|(p_\infty) = \lim_{k \rightarrow \infty} |A_k|(p_k) = 1, \quad (8.16)$$

Theorem 8.1	Chen--Li	rescaling	$ A_k = 1$
pointed compactness	(8.8)	fixed original-coordinate cutoff	
(8.13)--(8.14)	p	(8.7)	x_k $G_k(x_k, 0) = 0$
Gaussian		localized estimates	x_k
p			

9. fixed cutoff

transfer proof	fixed-center Proposition 5.1	ball inclusion	time translation	weighted
monotonicity		cutoff ϕ		
	$\phi(F) = \phi(X_0 + \lambda_j^{-1}(F_j + q_j)) .$			
$ F_j \leq R$	$ q_j \leq Q$	$\lambda_j^{-1}(F_j + q_j) \rightarrow 0$ uniformly	j	cutoff (3.2)
Gaussian weight	H_j, V_j	unweighted estimates (2.5)	f_j	monotonicity square (2.11)
shifted normal-position estimate		moving cutoff		

10.

1.	369	curvature-normalizing moving-center scaling	Han--Li--Sun	fixed-center tangent-flow
		scaling		
2.	“	Proposition 5.1	”	shrinking sphere ordinary MCF
3.	beta-symplectic flow			beta-symplectic
	q_j, θ_j	center-control hypotheses	fixed-center proof	moving-center version
	smooth nonflat normalized limit			
4.	q_j, θ_j	fixed-center Proposition 5.1		$q_j^\perp$ L^2
	$q_j \rightarrow 0$	local area bound		
5.	blow-up	fixed-center scaling	tangent behavior	curvature-normalizing
	moving-center scaling		nonflat singularity model	
6.	$F(x_j, t_j) \rightarrow X_0$	$q_j \rightarrow 0$	$\lambda_j \asymp (T - t_j)^{-1/2}$	sharp
	$ F(x_j, t_j) - X_0 = o(\sqrt{T - t_j})$			
7.	Chen--Li	4.7	q_k	q $q = 0$ x_k
	p	cutoff		

8.	pointed smooth compactness	localized monotonicity	$H_\infty = 0$
$(F_\infty + q)^\perp = 0$	$A_\infty = 0$		$q = 0$

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 2. X. Han, J. Li, and J. Sun, *Gradient flow for beta-symplectic critical surfaces*, Ann. Inst. H. Poincaré Anal. Non Linéaire 41 (2024), 1083--1116, especially Proposition 5.1. DOI: 10.4171/AIHPC/100.
 3. J. Chen and J. Li, *Mean curvature flow of surface in 4-manifolds*, Advances in Mathematics 163 (2001), 287--309, especially Theorem 4.7 on printed pp. 305--308. DOI: 10.1006/aima.2001.2008.